“EcoFleet AI: Intelligent Electric Vehicle Fleet Monitoring and Route Optimization Platform”

Agile Sprint Planning & User Story Workshop Report

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Agile Sprint Planning & User Story Workshop Report

ABSTRACT

The global shift toward electric mobility demands intelligent fleet management systems capable of handling EV-specific complexities such as battery health, charging optimization, and energy-efficient routing. This report presents an Agile Sprint Planning and User Story Workshop for EcoFleet AI, an AI-powered platform that monitors fleet performance, predicts battery degradation, optimizes charging schedules, and recommends energy-efficient routes in real time.

The workshop defines the project vision and stakeholders, decomposes requirements into Epics and User Stories, specifies measurable acceptance criteria in Given When Then format, prioritizes the product backlog, plans Sprint 1, estimates effort using Planning Poker with the Fibonacci sequence, and defines the Sprint Goal and expected deliverables demonstrating how Agile practices deliver incremental, testable, business-aligned value for a technically complex IoT/AI platform.

Keywords: Agile, Sprint Planning, User Stories, EV Fleet Management, IoT, Predictive Maintenance, Route Optimization, Scrum.

1. INTRODUCTION

The transportation sector is transitioning rapidly to electric vehicles (EVs) to cut emissions and fuel dependency, but managing EV fleets is more complex than managing conventional fleets requiring continuous tracking of battery health, charging schedules, energy consumption, and route efficiency. AI, IoT, GPS, and cloud computing now make it possible to monitor and predict fleet performance in real time.

Problem Statement: Conventional fleet systems track location and usage but lack intelligence to predict battery degradation, optimize charging, or minimize energy use through smarter routing. This causes unplanned downtime, inefficient charging costs, poor route planning, and limited real-time visibility for fleet managers.

Purpose: This report translates the EcoFleet AI problem statement into an actionable Agile plan covering stakeholder analysis, Epics/User Stories, acceptance criteria, backlog prioritization, sprint planning, and estimation ready for a Scrum team to begin development.

2. PROJECT VISION & STAKEHOLDERS

Vision: "To build an AI-powered, IoT-enabled EV Fleet Management Platform that maximizes vehicle uptime, extends battery life, and minimizes operational costs through predictive maintenance, intelligent charging, and energy-efficient routing accelerating sustainable, data-driven transportation."

Objectives: Monitor fleet performance via IoT; predict battery degradation using ML; optimize charging schedules; recommend energy-efficient routes; provide real-time dashboards; generate analytical reports; improve productivity while cutting costs and emissions.

Expected Outcomes: Longer battery life, higher fleet utilization, lower energy/charging costs, shorter travel times, increased vehicle availability, better decision-making, reduced emissions, and a scalable sustainable fleet platform.

Stakeholder Analysis:

Stakeholder	**Interest**
Fleet Manager	Primary user; monitors health, approves plans, makes operational decisions
Fleet Operators/Logistics Cos.	Business owner; concerned with ROI, uptime, cost savings
EV Drivers	Receive route/charging recommendations
Maintenance Technicians	Act on predictive maintenance alerts
Charging Station Operators	Provide station availability data
IoT/Telematics Provider	Supplies sensor data
Data Science/ML Team	Builds predictive models
System Administrator	Manages roles, security, configuration
Sustainability Officer	Tracks emissions/compliance
Product Owner	Prioritizes backlog
Scrum Master	Facilitates ceremonies
Development Team	Builds and delivers the platform

3. EPICS AND USER STORIES

Epic	Focus Area
E1	Real-Time Fleet & Battery Monitoring
E2	Predictive Maintenance
E3	Intelligent Charging Optimization
E4	Energy-Efficient Route Optimization
E5	Dashboards & Analytics
E6	User & Access Management

E1 – Real-Time Fleet & Battery Monitoring

- US1.1: As a fleet manager, I want real-time battery health status of all vehicles, so I can identify at-risk vehicles early.
- US1.2: As a fleet manager, I want a live vehicle location map, so I can track fleet activity.
- US1.3: As a fleet manager, I want alerts when battery health drops below a safe threshold, so I can act promptly.

E2 – Predictive Maintenance

- US2.1: As a fleet manager, I want predicted battery degradation trends, so I can plan replacements early.
- US2.2: As a technician, I want automated maintenance alerts from ML predictions, so I can service proactively.
- US2.3: As a fleet manager, I want a maintenance history log, so I can track recurring issues.

E3 – Intelligent Charging Optimization

- US3.1: As a fleet manager, I want optimal charging schedule recommendations, so I can reduce costs.
- US3.2: As an EV driver, I want real-time nearby charging station availability, so I can avoid delays.
- US3.3: As a fleet manager, I want charging prioritized for the lowest-battery vehicles, so availability is maximized.

E4 – Energy-Efficient Route Optimization

- US4.1: As a driver, I want the most energy-efficient route recommended, so I minimize energy use.
- US4.2: As a fleet manager, I want routes to factor in traffic and battery range, so unnecessary stops are avoided.

- US4.3: As a fleet manager, I want dynamic route reassignment on critically low battery, so delays are prevented.

E5 – Dashboards & Analytics

- US5.1: As a fleet manager, I want a real-time dashboard of fleet health/utilization/energy, so I can decide quickly.
- US5.2: As a sustainability officer, I want automated emissions/energy-savings reports, so I can track compliance.
- US5.3: As a fleet manager, I want to export analytical reports, so I can share insights with management.

E6 – User & Access Management

- US6.1: As an admin, I want to manage user roles/permissions, so data access stays secure.
- US6.2: As a fleet manager, I want secure login limited to my organization's data, so privacy is maintained.

4. ACCEPTANCE CRITERIA

US1.1: *Given* a logged-in fleet manager, *When* they open "Fleet Health," *Then* battery %, charge cycles, and status (Healthy/Warning/Critical) display for all vehicles, refreshed every 60 seconds.

US1.3: *Given* battery health falls below threshold, *When* sensor data is processed, *Then* an alert reaches the fleet manager within 2 minutes.

US2.1: *Given* historical battery data exists, *When* the ML model runs, *Then* a degradation forecast with remaining useful life and confidence score is generated.

US3.1: *Given* battery level, trip schedule, and tariff data, *When* a charging plan is requested, *Then* the system recommends cost-minimizing start/end times that meet the next trip deadline.

US4.1: *Given* a driver enters a destination, *When* the routing engine processes traffic/terrain/range, *Then* it returns the best route plus one alternate, with estimated energy use and arrival time.

US5.1: *Given* a fleet manager opens the dashboard, *When* data is available, *Then* total active vehicles, average battery health, energy used today, and vehicles under maintenance display together.

US6.2: *Given* a registered user logs in, *When* credentials are valid, *Then* access is granted strictly to their organization's/role's data.

5. PRODUCT BACKLOG PRIORITIZATION

Priority	User Story	Value	Dependency
1	US6.2 – Secure login	High	None
2	US1.1 – Battery monitoring	Critical	IoT integration
3	US1.3 – Low battery alerts	Critical	US1.1
4	US1.2 – Location map	High	GPS integration
5	US3.1 – Charging optimization	High	US1.1
6	US3.2 – Charging station availability	High	Station API
7	US4.1 – Route recommendation	High	GPS + traffic data
8	US2.1 – Degradation prediction	Med-High	ML model, history
9	US2.2 – Maintenance alerts	Medium	US2.1
10	US5.1 – Dashboard	Med-High	US1.1, US1.2, US3.1
11	US3.3 – Priority charging	Medium	US3.1
12	US4.2 – Traffic-aware routing	Medium	US4.1
13	US4.3 – Route reassignment	Medium	US4.1, US1.3
14	US2.3 – Maintenance log	Medium	US2.2
15	US5.2 – Compliance reports	Low-Med	US5.1
16	US5.3 – Report export	Low-Med	US5.1
17	US6.1 – Role management	Medium	US6.2

Rationale: Security and core monitoring come first since all other features depend on reliable authentication and live vehicle data. Charging and routing follow as the platform's differentiators; analytics/reporting come last since they consume data produced by earlier stories.

6. SPRINT BACKLOG

Sprint 1 targets the secure foundation and core monitoring, prerequisites for all later features.

User Story	Development Tasks
US6.2	Auth schema → Login API (JWT/OAuth) → Role-based access logic → Login UI → Tests
US1.1	IoT ingestion pipeline → Battery data model → Backend API → Dashboard UI → 60s refresh → Test with simulated data
US1.3	Threshold config → Alert-trigger service → Notification integration → Alert history UI → Latency testing
US1.2	GPS feed integration → Map component → Real-time updates → Multi-vehicle performance testing

7. STORY POINT ESTIMATION

Estimated via Planning Poker (Fibonacci: 1,2,3,5,8,13); disagreements resolved through discussion of technical unknowns (e.g., GPS accuracy).

User Story	Points	Justification
US6.2	5	Standard auth + moderate role-based access effort
US1.1	8	IoT + real-time pipeline + live UI = high complexity
US1.3	3	Builds on US1.1; simple alert logic
US1.2	5	GPS + map rendering = moderate complexity
US3.1	8	Multi-source optimization algorithm
US3.2	5	Third-party API integration
US4.1	8	Complex routing algorithm (traffic + terrain + range)
US2.1	13	Full ML model dev/training/validation; high uncertainty
US5.1	5	UI/aggregation, depends on prior APIs

Sprint 1 Capacity: 21 points (5+8+3+5)

8. SPRINT GOAL & EXPECTED DELIVERABLES

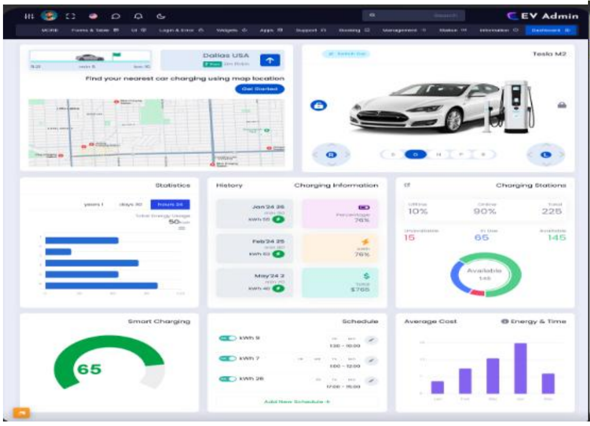
Sprint Goal: "Establish a secure, real-time fleet monitoring foundation authenticated access, live battery tracking, low-battery alerting, and live location visibility giving fleet managers immediate operational insight."

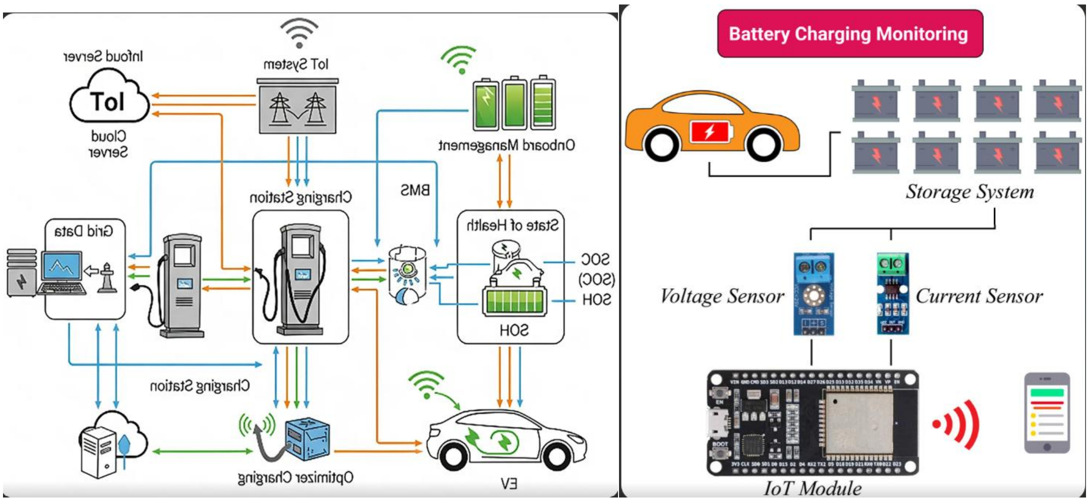
Deliverables: Secure role-based login; live Fleet Health dashboard (60s refresh); automated low-battery alerts (<2 min); real-time GPS location map; working IoT ingestion pipeline supporting future features; demo-ready increment (login → dashboard → alert → map).

Definition of Done: Acceptance criteria met and tested; code reviewed and deployed to staging; unit/integration tests passing; Product Owner acceptance at Sprint Review.

9. CONCLUSION

This workshop translates the EcoFleet AI problem statement into a structured, development-ready Agile plan. By defining the vision, mapping stakeholders, and decomposing requirements into testable Epics and User Stories, the plan keeps development tightly aligned with business value. The prioritized backlog, Sprint 1 plan, and story-point estimates give the team a clear, justified starting point and measurable delivery expectations establishing a strong foundation for iterative development across future sprints.





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